

Prithvi Chakravarthy

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EDUCATION

DePaul University

Master of Science in Computer Science

Chicago, IL

Sept. 2025 – Present

- Coursework: Distributed Systems, Applied Algorithms and Structures, DBMS, Real-Time System Architecture

BITS Pilani, WILP

Postgraduate Certificate in AI/ML

India

Oct. 2023 – Oct. 2024

- Coursework: Deep Learning, Neural Networks, Text Mining, Feature Engineering, Unsupervised Learning

VIT Bhopal

Bachelor of Technology in Computer Science

India

Jun. 2018 – Jun. 2022

- Coursework: Data Structures, Algorithms, DBMS, Operating Systems, Computer Architecture, Soft Computing, Computer Networks, Theory of Computation & Compiler Design

EXPERIENCE

Designare Solutions Pvt Ltd

Full Stack Developer

Mumbai, India

March 2022 – August 2025

- Developed and deployed full stack applications using React, Node.js, Python and PostgreSQL, improving scalability and performance.
- Designed distributed APIs and optimised data pipelines, reducing query time by 25%.
- Managed production deployments on Google Cloud, maintaining 99.9% uptime.
- Implemented secure user authentication and role-based access control (RBAC).
- Collaborated with stakeholders to define requirements and deliver on time.

PROJECTS

Space Invaders Game Engine | C#, Azul Engine, IrrKlang

- Engineered a high-performance Space Invaders clone using **C#** and the **Azul engine**, strictly adhering to **SOLID principles** and implementing **14+ GoF design patterns** to ensure a modular and decoupled architecture.
- Optimized memory management by architecting a universal **Object Pool** system (Template pattern), eliminating runtime heap allocation spikes and leveraging **Proxy/Flyweight patterns** to reduce GPU memory overhead for sprite rendering.
- Developed a decoupled collision detection system using **Visitor and Observer patterns** for double-dispatch resolution, and managed interval-based game events via a **Command pattern** paired with a **Priority Queue**.

Distributed File Retrieval & Indexing System | Java, gRPC, ZeroMQ, Protobuf

- Engineered a distributed search engine that indexes local files and retrieves file paths based on term frequency (TF) analysis across a multi-node network.
- Architected a dual-protocol communication layer using **gRPC** for synchronous file requests and **ZeroMQ** for broadcasting search queries, reducing retrieval latency by **45%**.
- Optimized data serialization with custom Protobuf schemas to handle high-frequency metadata exchanges, ensuring minimal overhead during large-scale file indexing operations.

Stable Diffusion Image Generation Model | Python, PyTorch, Jupyter

- Developed a latent diffusion model using PyTorch to generate high-fidelity images from text prompts, focusing on efficient tensor operations and model architecture.
- Implemented and optimized U-Net and attention mechanisms to improve inference speed and generation quality.
- Managed end-to-end data preprocessing and training pipelines, ensuring numerical stability and convergence across large datasets.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, Kotlin, C#

Frameworks/Libraries: React, Node.js, Pyramid, MaterialUI, TailwindCSS, PyTorch, TensorFlow

Databases: PostgreSQL, MySQL, Oracle

Tools: Google Cloud, GitHub, Docker, Jenkins, CI/CD, REST APIs, Jest